



LUND
UNIVERSITY

Written Examination
Linear Analysis, MATB24
October 27, 2021
Time: 14.00–19.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

You may freely use formulas from the formula sheet. Refer to relevant theorems from the course when you use them. Write clearly with concise motivations and explain your steps. In each of the six problems a maximum of five points can be achieved.

1. For $n \in \mathbb{N}$ let

$$f_n: [0, \infty) \rightarrow \mathbb{R}, \quad f_n(x) = \frac{2 + nx}{e^x + nx^2}.$$

- a) Prove that (f_n) converges pointwise on $[0, \infty)$ as $n \rightarrow \infty$ and determine the limit function.
- b) Prove that the convergence is uniform on $[1, 2]$, but not on $[0, 1]$.

Solution:

- a) $f_n(0) = 2$ for all $n \in \mathbb{N}$, so $\lim_{n \rightarrow \infty} f_n(x) = 2$. For $x > 0$ we have

$$f_n(x) = \frac{2/n + x}{e^x/n + x^2} \xrightarrow{n \rightarrow \infty} \frac{1}{x}.$$

Therefore, (f_n) converges pointwise to f , where

$$f(x) := \begin{cases} 2, & x = 0, \\ \frac{1}{x}, & x > 0. \end{cases}$$

- b) For all $x \in [1, 2]$:

$$|f_n(x) - f(x)| = \left| \frac{2x - e^x}{(e^x + nx^2)x} \right| \leq \frac{4 + e^2}{n} \xrightarrow{n \rightarrow \infty} 0.$$

Were the convergence uniform on $[0, 1]$, then f would have to be continuous on $[0, 1]$ because all f_n are continuous. But this is not the case.

Please, turn over!

2. Find a power series solution $u = u(x)$ to the problem

$$4xu'' + 6u' - u = 0, \quad u(0) = 1.$$

Also determine the radius of convergence of the obtained power series and write the solution in closed form for $x > 0$ and $x < 0$.

Solution: We make the ansatz $u(x) = \sum_{n=0}^{\infty} a_n x^n$ and formally differentiate under the sum. Then $a_0 \stackrel{!}{=} 1$ and

$$\begin{aligned} 0 &\stackrel{!}{=} \sum_{n=2}^{\infty} 4n(n-1)a_n x^{n-1} + \sum_{n=1}^{\infty} 6na_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n \\ &= \sum_{n=1}^{\infty} (2n(2n+1)a_n - a_{n-1})x^{n-1}, \end{aligned}$$

which yields the recurrence relation

$$a_0 = 1, \quad a_n = \frac{a_{n-1}}{2n(2n+1)}, \quad n \geq 1.$$

It is easy to see that this is solved by

$$a_n = \frac{1}{(2n+1)!}, \quad n \in \mathbb{N}_0.$$

Therefore, try

$$u(x) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^n.$$

Since by the above recurrence relation obviously

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}} = 0,$$

we conclude that the radius of convergence of the above power series is ∞ . Thus, u defined as above is smooth on $\mathbb{R}$ and all differentiations under the sum are justified; in particular, u is a solution of the problem. Moreover,

$$u(x) = \begin{cases} \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} \sqrt{x}^{2n+1} \sqrt{x}^{-1} = \frac{\sinh \sqrt{x}}{\sqrt{x}}, & x > 0, \\ \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \sqrt{-x}^{2n+1} \sqrt{-x}^{-1} = \frac{\sin \sqrt{-x}}{\sqrt{-x}}, & x < 0. \end{cases}$$

3. Let $f: \mathbb{R} \rightarrow \mathbb{C}$ be the 2π -periodic function that satisfies

$$f(t) = \begin{cases} (t + \pi)e^{it}, & -\pi < t < 0, \\ te^{it}, & 0 \leq t \leq \pi. \end{cases}$$

- a) Decide whether $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|$ is finite or infinite without using an explicit computation of the Fourier coefficients of f .
- b) Compute the (complex) Fourier series of f .
- c) Determine the value of $\sum_{n \in \mathbb{Z}} \hat{f}(n)$ if it exists.

Solution:

- a) Clearly, f is differentiable on $(-\pi, 0) \cup (0, \pi)$, so the Fourier inversion theorem yields that the Fourier series of f converges pointwise to f on $(-\pi, 0) \cup (0, \pi)$. Were $\sum_{n \in \mathbb{Z}} |\hat{f}(n)| < \infty$, then the Fourier series of f would be continuous on $\mathbb{T}$. However, f has a jump at 0, contradiction. Thus, necessarily $\sum_{n \in \mathbb{Z}} |\hat{f}(n)| = \infty$.
- b) We compute

$$\hat{f}(1) = \frac{1}{2\pi} \int_{-\pi}^{\pi} t dt + \frac{1}{2} \int_{-\pi}^0 dt = \frac{\pi}{2}$$

and for $n \in \mathbb{Z} \setminus \{1\}$

$$\begin{aligned} \hat{f}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} te^{-i(n-1)t} dt + \frac{1}{2} \int_{-\pi}^0 e^{-i(n-1)t} dt \\ &= \frac{(-1)^{n-1}i}{n-1} + \frac{i}{2(n-1)} - \frac{(-1)^{n-1}i}{2(n-1)} = \begin{cases} 0, & n \text{ even}, \\ \frac{i}{n-1}, & n \text{ odd}. \end{cases} \end{aligned}$$

Thus,

$$f(t) \sim \frac{\pi}{2} + \frac{i}{2} \sum_{k \in \mathbb{Z}, k \neq 0} \frac{e^{i(2k+1)t}}{k}.$$

- c) We have

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) = \frac{\pi}{2}.$$

This is obvious from setting $t = 0$ in the series above and also follows from the Fourier inversion theorem in view of $f(0-) = \pi$, $f(0+) = 0$.

Please, turn over!

4. Decide whether or not a function with the respective properties exists. If yes, provide an example; if not, prove that no such function can exist.
- a) $f \in L^1(\mathbb{T})$ with

$$\hat{f}(n) = \begin{cases} 27, & n = 20, \\ 10, & n = 21, \\ 0, & n \in \mathbb{Z} \setminus \{20, 21\}. \end{cases}$$

- b) $f \in L^1(\mathbb{T})$ with $\hat{f}(n) = 1$ for all $n \in \mathbb{Z}$.
- c) $f \in L^2(\mathbb{T})$ with $\hat{f}(n) = in^{-1/2}$ for all $n \in \mathbb{Z} \setminus \{0\}$.

Solution:

- a) Yes,

$$f(t) = 27e^{20it} + 10e^{21it}.$$

- b) No, necessarily $\lim_{|n| \rightarrow \infty} \hat{f}(n) = 0$ for $f \in L^1(\mathbb{T})$ by the Riemann–Lebesgue lemma.
- c) No, necessarily $\sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 < \infty$ for $f \in L^2(\mathbb{T})$ by Parseval.

5. Find a solution $u = u(x, t)$ for $(x, t) \in [0, \pi/2] \times [0, \infty)$ to the following problem for the heat equation:

$$\begin{cases} u_t = 2u_{xx}, & 0 < x < \pi/2, t > 0, \\ u(0, t) = u_x(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = x(\pi - x), & 0 < x < \pi/2. \end{cases}$$

Also briefly discuss the regularity of your solution and in which sense the initial condition is satisfied for it.

Solution: We follow the method of separation of variables and begin by inserting the ansatz $u(x, t) = X(x)T(t)$ ($X \neq 0$, $T \neq 0$) into the heat equation. This yields

$$XT' = 2X''T \Rightarrow X'' = \lambda X, T' = 2\lambda T$$

for some constant $\lambda \in \mathbb{R}$. Equipping the equation for X with the boundary conditions $X(0) = 0$, $X'(\pi/2) = 0$, we can only get nontrivial solutions for $\lambda < 0$, namely

$$X(x) = a \sin(\sqrt{-\lambda}x) + b \cos(\sqrt{-\lambda}x)$$

for some constants $a, b \in \mathbb{R}$. Due to the boundary condition $X(0) = 0$, necessarily $b = 0$, and by the other boundary condition $X'(\pi/2) = 0$ we obtain a nontrivial solution if and only if

$$\cos(\sqrt{-\lambda}\pi/2) = 0 \Leftrightarrow \lambda = -(2n+1)^2, n \in \mathbb{N}_0.$$

To sum up, we obtain $X_n(x) = a_n \sin((2n+1)x)$ for some constant $a_n \in \mathbb{R}$ and where $n \in \mathbb{N}_0$. On the other hand, the Cauchy problem $T'_n = -2(2n+1)^2 T_n$, $T_n(0) = 1$ (say) has the unique solution $T_n(t) = e^{-2(2n+1)^2 t}$, where $n \in \mathbb{N}_0$. Therefore, we get the solutions $u_n(x, t) = a_n e^{-2(2n+1)^2 t} \sin((2n+1)x)$ (of the heat equation together with the boundary conditions), $n \in \mathbb{N}_0$, and superposing them yields (formally) the trial solution

$$u(x, t) = \sum_{n=0}^{\infty} a_n e^{-2(2n+1)^2 t} \sin((2n+1)x).$$

Let us now define $f(x) := x(\pi - x)$, $x \in [0, \pi]$, and compute the sine series of f . The corresponding coefficients b_n , $n \in \mathbb{N}$, are given by

$$\begin{aligned} b_n &= 2 \int_0^{\pi} x \sin(nx) dx - \frac{2}{\pi} \int_0^{\pi} x^2 \sin(nx) dx \\ &= -\frac{2\pi(-1)^n}{n} + \frac{2\pi(-1)^n}{n} - \frac{4\pi}{n} \int_0^{\pi} x \cos(nx) dx = \frac{4\pi}{n^2} \int_0^{\pi} \sin(nx) dx \\ &= \begin{cases} 0, & n \text{ even}, \\ \frac{8}{\pi n^3}, & n \text{ odd}. \end{cases} \end{aligned}$$

Therefore, choose $a_n := b_{2n+1}$, $n \in \mathbb{N}_0$; thus,

$$u(x, t) = \frac{8}{\pi} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} e^{-2(2n+1)^2 t} \sin((2n+1)x).$$

Because of the exponential decay of $e^{-2(2n+1)^2 t}$ in n for $t > 0$ and by $\sum_{n=1}^{\infty} |b_n| < \infty$ we have $u \in C^\infty(\mathbb{R} \times (0, \infty)) \cap C(\mathbb{R} \times [0, \infty))$, $u(x, 0) = x(\pi - x)$ for $0 < x < \pi/2$, and $u(\cdot, t) \rightarrow f$ uniformly on $[0, \pi/2]$ as $t \searrow 0$. (Not necessary: Since in fact $\sum_{n=1}^{\infty} n|b_n| < \infty$, we even have $u \in C^1(\mathbb{R} \times [0, \infty))$ and additionally $u_x(\cdot, t) \rightarrow f'$ uniformly on $[0, \pi/2]$ as $t \searrow 0$.)

Please, turn over!

6. Let $f, g: \mathbb{R} \rightarrow \mathbb{C}$,

$$f(t) = |t|e^{-|t|}, \quad g(t) = \begin{cases} 0, & t < 0, \\ e^{-2t}, & t \geq 0. \end{cases}$$

- a) Compute the Fourier transforms of f and g .
b) Use the results to determine the integrals

$$\int_{-\infty}^{\infty} \frac{(1-x^2)^2}{(1+x^2)^4} dx \quad \text{and} \quad \int_{-\infty}^{\infty} \frac{1-x^2}{(1+x^2)^2(4+x^2)} dx.$$

Solution:

- a) We compute for $\omega \in \mathbb{R}$

$$\begin{aligned} \hat{f}(\omega) &\stackrel{f \text{ even}}{=} 2 \operatorname{Re} \int_0^{\infty} t e^{(-1-i\omega)t} dt = -2 \operatorname{Re} \left(\frac{1}{-1-i\omega} \int_0^{\infty} e^{(-1-i\omega)t} dt \right) \\ &= 2 \operatorname{Re} \frac{1}{(1+i\omega)^2} = \frac{2(1-\omega^2)}{(1+\omega^2)^2} \end{aligned}$$

and

$$\hat{g}(\omega) = \int_0^{\infty} e^{(-2-i\omega)t} dt = \frac{1}{2+i\omega} = \frac{2-i\omega}{4+\omega^2}.$$

- b) We have

$$|\hat{f}(\omega)|^2 = \frac{4(1-\omega^2)^2}{(1+\omega^2)^4}, \quad \hat{f}(\omega)\overline{\hat{g}(\omega)} = \frac{2(1-\omega^2)(2+i\omega)}{(1+\omega^2)^2(4+\omega^2)},$$

so by Plancherel's formula

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{(1-x^2)^2}{(1+x^2)^4} dx &= \frac{1}{4} \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 d\omega = \frac{\pi}{2} \int_{-\infty}^{\infty} |f(t)|^2 dt = \pi \int_0^{\infty} t^2 e^{-2t} dt \\ &= \pi \int_0^{\infty} t e^{-2t} dt = \frac{\pi}{2} \int_0^{\infty} e^{-2t} dt = \frac{\pi}{4} \end{aligned}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{1-x^2}{(1+x^2)^2(4+x^2)} dx &= \frac{1}{4} \operatorname{Re} \int_{-\infty}^{\infty} \hat{f}(\omega)\overline{\hat{g}(\omega)} d\omega = \frac{\pi}{2} \operatorname{Re} \int_{-\infty}^{\infty} f(t)\overline{g(t)} dt \\ &= \frac{\pi}{2} \int_0^{\infty} t e^{-3t} dt = \frac{\pi}{6} \int_0^{\infty} e^{-3t} dt = \frac{\pi}{18}. \end{aligned}$$